Session 2 – Self-bounding learning with random forests

Goal. By the end of the session you should be able to learn the weights of a forest by minimizing a certificate, compare relaxed and exact objectives, relate strength and diversity to the certificates, and select a model with certificates instead of cross-validation.
To reread before the session. Sections 5, 6 and 7.2 of the notes; notebook 4.
Timing (3 hours). Part 2.1 60 min
Part 2.2 45 min
Part 2.3 30 min
Part 2.4 45 min
To hand in at the end of the session. Your notebook, with the table of Part 2.1 with means and standard deviations, and the model-selection table of Part 2.4.

Part 2.1 – Learning the weights (Sections 6 and 7.2)

1. On each dataset, compute the posteriors minimizing the first-order bound (`optimize_fo`) and the tandem bound (`optimize_tnd`). Report the test risk of the corresponding votes, their certificates and the *effective number of trees* $1/\sum_t \rho_t^2$.
2. Produce a table with one row per dataset and, for each posterior (uniform, FO, TND), the mean and standard deviation over the 5 splits of the test risk and of the certificate. For the uniform posterior, report the best of the three certificates: what correction does this choice require?
3. Which posterior gives the best vote? the best certificate? Are the conclusions of notebook 4 confirmed?

Part 2.2 – Minimizing the kl certificate directly (Section 6.6)

The optimizers of Part 2.1 minimize a *relaxation* of the certificate (PAC-Bayes-λ). Since the certificate is valid for any posterior, we can also minimize the kl-form certificate itself.

1. Using `certificate_and_grad` (value and gradient of $4\,\overline{\mathrm{kl}}^{-1}(\rho^\top \widehat{\mathbf{L}}\rho, \varepsilon(\rho))$ with respect to the softmax logits, Proposition 6.6), implement Adam (300 iterations, step 0.05, starting from the uniform posterior).
2. On `spambase`, `satimage`, `wdbc` and `pageblocks`, compare the uniform posterior, the TND-optimized posterior of Part 2.1 and the posterior found by Adam: certificate, test risk, effective number of trees. Plot the certificate along the iterations.
3. Does the choice of the optimizer, of the step size or of the number of iterations require a union bound? Why?

Part 2.3 – Strength versus diversity (Section 5)

1. On `spambase` and `german`, learn forests with `max_features` $\in \{1, \sqrt{d}, d/2, d\}$ (3 splits). Report the test Gibbs risk, the test disagreement, the test risk of the vote and the OOB tandem and C-bound certificates.
2. Describe how strength and diversity evolve. Which value gives the best vote? Do the certificates follow the risk of the vote?

Part 2.4 – Self-bounding model selection (Section 6)

1. On `spambase`, `pima`, `wdbc` and `german`, learn $K = 5$ forests with `max_depth` $\in \{1, 2, 4, 8, \mathrm{None}\}$, minimize the tandem bound of each of them and compute its certificate with δ/K. Select the depth with the smallest certificate.

2. Compare with the depth selected by 5-fold cross-validation of the uniform forest on the training set, and with the best depth in hindsight (smallest test risk). Report the test risks of the selected models.
3. Why must the certificates be computed with δ/K? When does this selection rule fail, and what should be done then?